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WSD - Lab Exercise 6

Intent behind using this design:

1. Design Philosophy and User Experience:
 - Simplicity and Professional Appearance: Clean, modern look achieved with Tailwind CSS for easy navigation.
 - Responsiveness: Tailwind ensures form adapts to various screen sizes, providing consistent user experience.
 - User Feedback: Real-time feedback with green checkmarks for valid inputs and red "x" for invalid inputs.
2. Form Structure and Layout:
 - Centered Layout: Draws focus and makes the form approachable with padding, rounded corners, and shadows.
 - Field Grouping: Organized, easy-to-read layout with clear labels and evenly spaced input fields.
3. Implementation of Input Validation:
 - JavaScript Validation: Real-time feedback as the user types, preventing form submission errors.
 - Error Messaging: Red error messages appear below fields when validation fails, providing clear guidance.
4. Validation Logic and Criteria:
 - Name Validation: Accepts alphabetic characters and spaces, minimum 3 characters.
 - Rationale: Prevents common input errors for names.
 - Email Validation: Ensures proper email format (e.g., user@example.com).

- Rationale: Prevents submission of invalid email addresses.
- Password Validation: At least 8 characters, containing both letters and numbers.
 - Rationale: Ensures basic password security requirements.
- Confirm Password Validation: Matches password field to confirm user input accuracy.
- DOB Validation: Ensures users are at least 18 years old with date format "YYYY-MM-DD".
 - Rationale: Complies with common age restrictions for user registrations.

5. Real-time Feedback and Visual Indicators:

- Dynamic Icons: ✓ for valid and ✗ for invalid states, providing intuitive feedback.
- Border Color Changes: Green borders for valid and red for invalid inputs to enhance accessibility.

6. Use of Tailwind CSS:

- Utility Classes: Simplifies styling and makes it easy to maintain the form's design.
- Responsive Design: Ensures the form looks good on all devices.
- Consistent Design Language: Maintains a cohesive look and feel across form elements.

7. Accessibility Considerations:

- Labels and Error Messaging: Improves accessibility, ensuring screen readers identify fields correctly.
- Color Contrast: High contrast used for error messages/icons for readability by users with visual impairments.

8. Form Submission Handling:

- Prevent Default Behavior: Ensures no data is submitted until all fields pass validation.
- Disabled Submit Button: Only enabled when all validation criteria are met to prevent incomplete submissions.

HTML Code with Proper Documentation :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration Form</title>
  <!-- Including Tailwind Css here -->
  <script src="https://cdn.tailwindcss.com"></script>
  <style>
    .icon {
      position: absolute;
      right: 10px;
      top: 12px;
      font-size: 18px;
      display: none;
    }

    .icon.valid {
      color: green;
      display: block;
    }

    .icon.invalid {
      color: red;
      display: block;
    }
  </style>
</head>

<!-- all structure elements along with tailwind classes for easier beautification -->
<!-- adding additional elements to display errors, invalid inputs and
tickmarks/crosses -->
<body class="bg-gray-100 min-h-screen flex items-center justify-center p-4">
```

```

<div class="bg-white rounded-lg shadow-lg p-8 w-full max-w-md">
  <h2 class="text-2xl font-bold text-center text-gray-800 mb-6">User
Registration Form</h2>
  <form id="registrationForm" class="space-y-5">
    <div class="form-group relative">
      <label for="name" class="block text-sm font-medium
text-gray-700">Name:</label>
      <input type="text" id="name" name="name" required
      class="mt-1 block w-full px-3 py-2 bg-white border border-gray-300
rounded-md shadow-sm focus:outline-none focus:ring-indigo-500
focus:border-indigo-500 sm:text-sm">
      <span class="icon" id="nameIcon">✓</span>
      <span class="error-message text-red-500 text-xs mt-1"
id="nameError">Name must be at least 3 characters long and contain only letters
and spaces.</span>
    </div>

    <div class="form-group relative">
      <label for="email" class="block text-sm font-medium
text-gray-700">Email:</label>
      <input type="email" id="email" name="email" required
      class="mt-1 block w-full px-3 py-2 bg-white border border-gray-300
rounded-md shadow-sm focus:outline-none focus:ring-indigo-500
focus:border-indigo-500 sm:text-sm">
      <span class="icon" id="emailIcon">✓</span>
      <span class="error-message text-red-500 text-xs mt-1"
id="emailError">Please enter a valid email address.</span>
    </div>

    <div class="form-group relative">
      <label for="password" class="block text-sm font-medium
text-gray-700">Password:</label>
      <input type="password" id="password" name="password" required
      class="mt-1 block w-full px-3 py-2 bg-white border border-gray-300
rounded-md shadow-sm focus:outline-none focus:ring-indigo-500
focus:border-indigo-500 sm:text-sm">

```

```
        <span class="icon" id="passwordIcon">✓</span>
        <span class="error-message text-red-500 text-xs mt-1"
id="passwordError">Password must be at least 8 characters long and contain
both letters and numbers.</span>
    </div>
```

```
    <div class="form-group relative">
        <label for="confirmPassword" class="block text-sm font-medium
text-gray-700">Confirm Password:</label>
        <input type="password" id="confirmPassword"
name="confirmPassword" required
        class="mt-1 block w-full px-3 py-2 bg-white border border-gray-300
rounded-md shadow-sm focus:outline-none focus:ring-indigo-500
focus:border-indigo-500 sm:text-sm">
        <span class="icon" id="confirmPasswordIcon">✓</span>
        <span class="error-message text-red-500 text-xs mt-1"
id="confirmPasswordError">Passwords do not match.</span>
    </div>
```

```
    <div class="form-group relative">
        <label for="dob" class="block text-sm font-medium text-gray-700">Date
of Birth:</label>
        <input type="date" id="dob" name="dob" required
        class="mt-1 block w-full px-3 py-2 bg-white border border-gray-300
rounded-md shadow-sm focus:outline-none focus:ring-indigo-500
focus:border-indigo-500 sm:text-sm">
        <span class="icon" id="dobIcon">✓</span>
        <span class="error-message text-red-500 text-xs mt-1"
id="dobError">You must be at least 18 years old.</span>
    </div>
```

```
        <button type="submit" class="w-full bg-indigo-600 text-white font-medium
py-2 rounded-md shadow-sm hover:bg-indigo-700 focus:outline-none
focus:ring-2 focus:ring-offset-2 focus:ring-indigo-500 disabled:opacity-50"
id="submitBtn" disabled>Register</button>
    </form>
</div>
```

```

<script>
    // instantiating all ids and variables
    const form = document.getElementById('registrationForm');
    const nameInput = document.getElementById('name');
    const emailInput = document.getElementById('email');
    const passwordInput = document.getElementById('password');
    const confirmPasswordInput =
document.getElementById('confirmPassword');
    const dobInput = document.getElementById('dob');
    const submitBtn = document.getElementById('submitBtn');

    const nameError = document.getElementById('nameError');
    const emailError = document.getElementById('emailError');
    const passwordError = document.getElementById('passwordError');
    const confirmPasswordError =
document.getElementById('confirmPasswordError');
    const dobError = document.getElementById('dobError');

    const nameIcon = document.getElementById('nameIcon');
    const emailIcon = document.getElementById('emailIcon');
    const passwordIcon = document.getElementById('passwordIcon');
    const confirmPasswordIcon =
document.getElementById('confirmPasswordIcon');
    const dobIcon = document.getElementById('dobIcon');

    //All Validation Logic:
    //If conditions satisfied, adding success classes else keeping failed classes
    by default.

    //Name Validation
    function validateName() {
        const namePattern = /^[A-Za-z\s]{3,}$/;
        if (namePattern.test(nameInput.value)) {
            nameInput.classList.add('border-green-500');
            nameInput.classList.remove('border-red-500');
            nameError.style.display = 'none';

```

```

        nameIcon.classList.add('valid');
        nameIcon.classList.remove('invalid');
        nameIcon.textContent = '✓';
    } else {
        nameInput.classList.remove('border-green-500');
        nameInput.classList.add('border-red-500');
        nameError.style.display = 'block';
        nameIcon.classList.remove('valid');
        nameIcon.classList.add('invalid');
        nameIcon.textContent = '✗';
    }
}

```

//Email Validation

```

function validateEmail() {
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (emailPattern.test(emailInput.value)) {
        emailInput.classList.add('border-green-500');
        emailInput.classList.remove('border-red-500');
        emailError.style.display = 'none';
        emailIcon.classList.add('valid');
        emailIcon.classList.remove('invalid');
        emailIcon.textContent = '✓';
    } else {
        emailInput.classList.remove('border-green-500');
        emailInput.classList.add('border-red-500');
        emailError.style.display = 'block';
        emailIcon.classList.remove('valid');
        emailIcon.classList.add('invalid');
        emailIcon.textContent = '✗';
    }
}

```

//Password Validation

```

function validatePassword() {
    const passwordPattern = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d]{8,}$/;
    if (passwordPattern.test(passwordInput.value)) {

```

```

        passwordInput.classList.add('border-green-500');
        passwordInput.classList.remove('border-red-500');
        passwordError.style.display = 'none';
        passwordIcon.classList.add('valid');
        passwordIcon.classList.remove('invalid');
        passwordIcon.textContent = '✓';
    } else {
        passwordInput.classList.remove('border-green-500');
        passwordInput.classList.add('border-red-500');
        passwordError.style.display = 'block';
        passwordIcon.classList.remove('valid');
        passwordIcon.classList.add('invalid');
        passwordIcon.textContent = '✗';
    }
    validateConfirmPassword();
}

//Confirm Password Validation
function validateConfirmPassword() {
    if (confirmPasswordInput.value === passwordInput.value &&
confirmPasswordInput.value !== "") {
        confirmPasswordInput.classList.add('border-green-500');
        confirmPasswordInput.classList.remove('border-red-500');
        confirmPasswordError.style.display = 'none';
        confirmPasswordIcon.classList.add('valid');
        confirmPasswordIcon.classList.remove('invalid');
        confirmPasswordIcon.textContent = '✓';
    } else {
        confirmPasswordInput.classList.remove('border-green-500');
        confirmPasswordInput.classList.add('border-red-500');
        confirmPasswordError.style.display = 'block';
        confirmPasswordIcon.classList.remove('valid');
        confirmPasswordIcon.classList.add('invalid');
        confirmPasswordIcon.textContent = '✗';
    }
}

```

//DOB Validation

```
function validateDOB() {  
    const dobValue = new Date(dobInput.value);  
    const today = new Date();  
    const age = today.getFullYear() - dobValue.getFullYear();  
    const ageCheck = today.setFullYear(today.getFullYear() - 18);  
  
    if (dobInput.value && dobValue <= new Date(ageCheck)) {  
        dobInput.classList.add('border-green-500');  
        dobInput.classList.remove('border-red-500');  
        dobError.style.display = 'none';  
        doblcon.classList.add('valid');  
        doblcon.classList.remove('invalid');  
        doblcon.textContent = '✓';  
    } else {  
        dobInput.classList.remove('border-green-500');  
        dobInput.classList.add('border-red-500');  
        dobError.style.display = 'block';  
        doblcon.classList.remove('valid');  
        doblcon.classList.add('invalid');  
        doblcon.textContent = '✗';  
    }  
}
```

//Total validity (Checking for overall validation, and enabling submit button if yes)

```
function checkFormValidity() {  
    if (  
        nameInput.classList.contains('border-green-500') &&  
        emailInput.classList.contains('border-green-500') &&  
        passwordInput.classList.contains('border-green-500') &&  
        confirmPasswordInput.classList.contains('border-green-500') &&  
        dobInput.classList.contains('border-green-500')  
    ) {  
        submitBtn.disabled = false;  
    } else {  
        submitBtn.disabled = true;  
    }  
}
```

```

    }
}

//Adding event listeners to check for user inputs and form changes
nameInput.addEventListener('input', () => {
    validateName();
    checkFormValidity();
});

emailInput.addEventListener('input', () => {
    validateEmail();
    checkFormValidity();
});

passwordInput.addEventListener('input', () => {
    validatePassword();
    checkFormValidity();
});

confirmPasswordInput.addEventListener('input', () => {
    validateConfirmPassword();
    checkFormValidity();
});

dobInput.addEventListener('input', () => {
    validateDOB();
    checkFormValidity();
});

form.addEventListener('submit', (e) => {
    e.preventDefault();
    alert('Form submitted successfully!');
});
</script>
</body>
</html>

```

Screenshots:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5500/LabExercise-6/withtailwind.html'. The main content is a 'User Registration Form' centered on a light gray background. The form has a white background and a thin gray border. It contains the following fields and messages:

- Name:** An empty text input field. Below it, a red error message reads: 'Name must be at least 3 characters long and contain only letters and spaces.'
- Email:** An empty text input field. Below it, a red error message reads: 'Please enter a valid email address.'
- Password:** An empty text input field. Below it, a red error message reads: 'Password must be at least 8 characters long and contain both letters and numbers.'
- Confirm Password:** An empty text input field. Below it, a red error message reads: 'Passwords do not match.'
- Date of Birth:** A date picker input field showing 'dd-mm-yyyy'. Below it, a red error message reads: 'You must be at least 18 years old.'
- Register:** A blue button with white text.

The screenshot shows the same 'User Registration Form' as the first image, but now it is filled with data. Each input field has a green checkmark icon on the right side, indicating successful validation. The data entered is:

- Name:** 'Karan Agarwal'
- Email:** 'karan@gmail.com'
- Password:** '*****'
- Confirm Password:** '*****'
- Date of Birth:** '20-09-2003'
- Register:** A blue button with white text.

127.0.0.1:5500 says
Form submitted successfully!

OK

User Registration Form

Name:

Karan Agarwal

Email:

karan@gmail.com

Password:

Confirm Password:

Date of Birth:

20-09-2003

Register